Similar Lengths

Exam code: 0607

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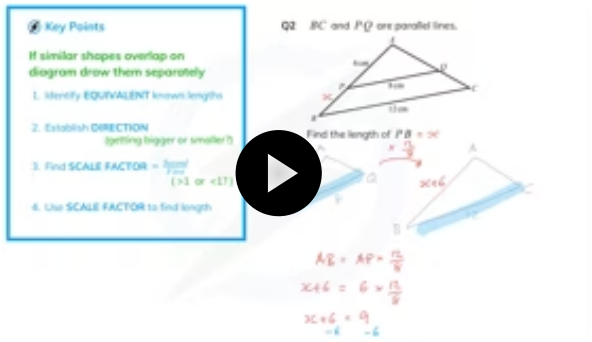

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Similar lengths

How do I find the scale factor between lengths on similar shapes?

- Equivalent **lengths** on two similar shapes will be in the same ratio and are linked by a **scale factor**
- Establish the **type** of enlargement
  - If the second shape is **bigger**
    - then the **scale factor** is **greater than 1**
  - If the second shape is **smaller**
    - then the **scale factor** is **greater than 0** but **less than 1**
- To find the **scale factor**
  - Identify **known** lengths of **corresponding sides**
  - **Divide** the length on the **second shape** by the corresponding length on the **first shape**

How do I find missing lengths on similar shapes?

Method 1

- STEP 1
  Find the **scale factor** to get from the **first** shape to the **second** shape

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- STEP 2
  Use the scale factor to find the **length** you need
  - To find a missing length on the **second shape**
    - **Multiply** the corresponding length on the first shape by the scale factor
  - To find a missing length on the **first shape**
    - **Divide** the corresponding length on the first shape by the scale factor

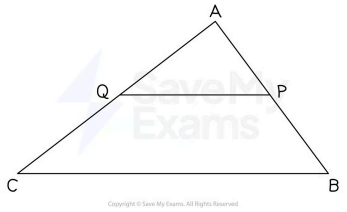
Method 2

- STEP 1
  Find the **scale factor** to get from the **smaller** shape to the **bigger** shape
  - **Divide** a length on the **bigger** shape by the corresponding length on the **smaller** shape
  - The scale factor is **always greater than 1** for this method
- STEP 2
  Use the scale factor to find the **length** you need
  - To find a missing length on the **bigger shape**
    - **Multiply** the corresponding length on the smaller shape by the scale factor
  - To find a missing length on the **smaller shape**
    - **Divide** the corresponding length on the bigger shape by the scale factor

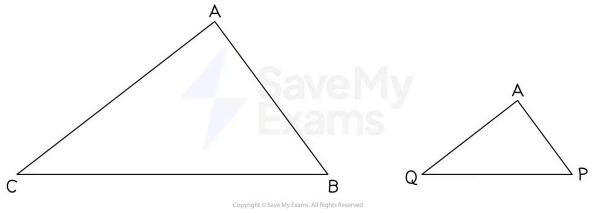
Examiner Tips and Tricks

If similar shapes overlap on the diagram (or are not clear) draw them separately.

For example, in this diagram the triangles ABC and APQ are similar:

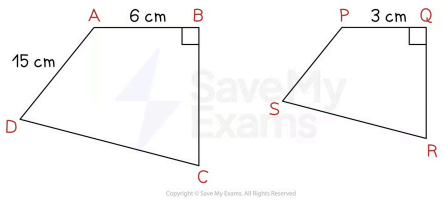


So redraw them separately before starting:



Worked Example

ABCD and *PQRS* are similar shapes.
Find the length of *PS*.



Answer:

The two shapes are mathematically similar

Identify two known corresponding sides of the similar shapes

AB and PQ are corresponding sides

Method 1

The second shape is smaller than the first shape so the scale factor will be between 0 and 1
Divide the known length on the second shape by the corresponding length on the first shape to find the scale factor

$$\text{Scale Factor} = \frac{3}{6} = \frac{1}{2}$$

Multiply the length AD by the scale factor to find its corresponding length PS on the second shape

$$PS = \frac{1}{2} \times 15 = \frac{15}{2}$$

$$PS = 7.5 \text{ cm}$$

Method 2

Divide the known length on the bigger shape by the corresponding length on the smaller shape to find the scale factor

$$\text{Scale Factor} = \frac{6}{3} = 2$$

The missing length is on the smaller shape

Divide the length AD by the scale factor to find its corresponding length PS on the smaller shape

$$PS = 15 \div 2$$

$$PS = 7.5 \text{ cm}$$

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